Alessandra Giorgi and Erika Petrocchi*

Ca' Foscari University of Venice

August 2025

Preliminary version.

Final version:
Alessandra Giorgi, Erika Petrocchi, On the syntactic representation of co-speech gestures in expressive language, in "Lingue e linguaggio" 1/2025, pp. 21-39,
doi: 10.1418/117441 URL: https://www.rivisteweb.it/doi/10.1418/117441

On the syntactic representation of co-speech gestures in expressive language

1 Introduction

In recent years, the relationship between gestures and linguistic expression has gained increasing attention in the field of linguistics. Gestures are now recognized as a crucial part of language, which is widely considered to be inherently multimodal, where by *multimodality in language*, we refer to the way linguistic expression is conveyed through both vocal and visual channels, utilizing prosody, manual and non-manual gestures and the positioning of the body in space. Traditionally, gestures have been studied from a semantic perspective (e.g., Ebert & Ebert, 2014; Schlenker, 2018; Esipova, 2019). However, scholars have also begun to examine them as elements integrated into grammatical structure (Jouitteau, 2004, 2007; Giorgi, 2015, 2018; Giorgi & Dal Farra, 2019; Ippolito, 2019, 2021; Sailor & Colasanti, 2020; Colasanti, 2021a, 2021b, 2023; Ippolito et al., 2022; Giorgi & Petrocchi, 2024).

In this paper, we present an integrated perspective on the relationship between syntax and pragmatics. Our main thesis is that certain phenomena in expressive language can be better understood and explained through a model that combines syntax, prosody, and gesture. The interplay among these components can convey emotional meaning effectively.

We consider here co-speech gestures within emotional contexts, in particular, surprise and disapproval, as we will further clarify in section 3, across various languages belonging to different language families.

* Erika Petrocchi is supported by the TheNextGeneratonEU and the Italian Ministry of University and Research (MUR) as part of the PRIN Project CUP: H53D23004410006

Specifically, we are going to address three key questions: How are gestures triggered? To what extent are they language-specific – that is, how universal might they be? And what role do they play in language.

2 Theoretical background

The minimalist program, as proposed by Chomsky (1995, 2000, 2001, 2008) and other scholars, hypothesizes the existence of a representational level, i.e., syntax – more accurately, *core syntax*. Core syntax connects to the phonological system through the *sensorimotor component* and to the interpretive system via the *conceptual component*. In other words, syntax serves as the link between sound and meaning. This relationship is illustrated in the following figure:[1]

The Minimalist model:

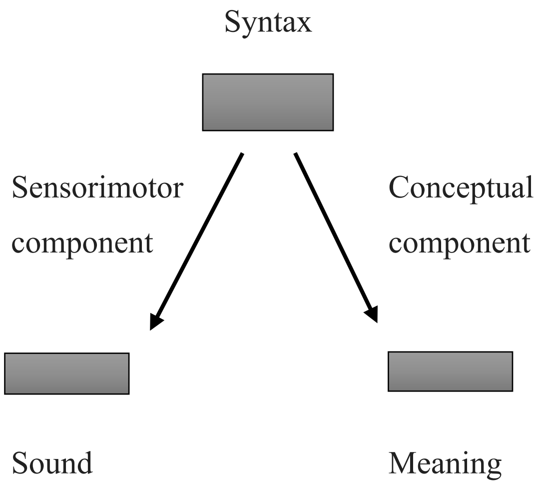


Figure 1

In this model, the only way to connect sound and meaning is through the representational level. Let us briefly consider the history of this crucial intuition. European structuralism, most notably De Saussure (1916), introduced the idea that the relationship between sound and meaning is arbitrary.[2] The generative proposal, which began with Chomsky (1957), incorporated this basic notion and consistently focused on an underlying level of linguistic

[1] This diagram visually represents the achitecture of the generative Minimalist model. For a recent discussion of the relations among the various components, see Chomsky et al. (2019).

[2] Many scholars have challenged this traditional view about arbitrariness. In recent times, experimental evidence has been presented – see works by D'Anselmo et al. (2019), Kantartzis et al. (20011) and Imai et al. (2008), among others – that suggests that there are non-random relationships between specific sounds and their associated meanings and that this might be particularly relevant to the process of language acquisition. Regardless of the perspective on the relationship between the two, it is essential to note that in these works, no reference is made to a representational level.

representation. This concept evolved further in Chomsky's (1975), where the Transformational Grammar framework introduced the idea that Deep Structure serves as the basis for deriving the actual form of a sentence. This process results in the correct pairing of sound and meaning, represented respectively by Surface Structure and Logical Form.

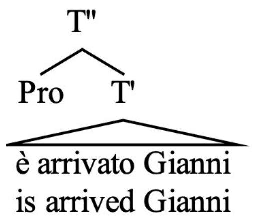
Importantly, as already pointed out above, in this generative model there is no direct link between sound and meaning, in the sense that the representational level, i.e., core syntax, is always mediating between the two.

To exemplify, capitalizing on the discussion in Giorgi (2023), let's consider a simple case of yes-no question in Italian, with a post-verbal subject, a possibility available to pro-drop languages, as the following one:

(1) E' arrivato Gianni
 lit: is arrived Gianni
 Gianni arrived
 Did Gianni arrive?

This sentence can be associated with two intonations, each corresponding to a different interpretation: the interrogative intonation, giving rise to a yes-no question, and the affirmative intonation, giving rise to an assertion.[3] One might conclude, therefore, that prosody directly disambiguates the two meanings. However, in generative grammar, since the development of the transformational approach (Chomsky, 1975), the two meanings, and consequently the two phonological realizations, are associated with different syntactic representations. Simplifying a complex technical and theoretical discussion, it is possible to assign the following structure to the affirmative sentence:

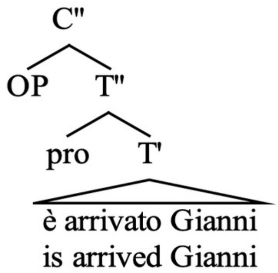
(2)



Whereas the interrogative one is represented as follows:

[3] For an analysis of intonation in Italian, see, among others, Avesani (1995) and Grice et al. (2005). For reasons of space, we are not going to elaborate further on this point.

(3)



In the interrogative structure, a non-lexical operator is present. This hypothesis, i.e., the presence of an empty operator in yes-no questions, has several important empirical consequences and can explain many phenomena in several languages, showing that it is not an *ad hoc* hypothesis to be adopted just for these cases.

From this example, it is clear how syntax mediates the relationship between sound and meaning: The empty operator creates the relevant difference between the two structures and can thus be seen as the trigger for the prosodic realization on one hand and the intended interpretation on the other. These silent heads triggering a specific sensorimotor realization are referred to as *prosody-oriented heads* (Giorgi 2016) and the analysis has been extended to various cases, such as parentheticals and Italian clitic left dislocation (CLLD), as discussed in Giorgi (2019).

It is important to note that the model described above is less powerful and consequently more restrictive than one *also* including a direct relationship between sound and meaning. A more restrictive model is easier to falsify, making it the preferable *null hypothesis* to be considered first. Only when it is proven that the null hypothesis is not viable can one move to a more powerful, less restrictive model.

Therefore, our theoretical analysis of co-speech gestures takes the consideration above as a starting point: we submit that the sensorimotor realization is triggered by syntactic cues and results in a multimodal realization of the sentence associated with the special emotional interpretation. In what follows we are going to illustrate some examples to this end.

More recent experimental studies within formal linguistics, particularly in the minimalist framework, have focused on evidence that syntax, prosody, and gestures systematically align. This alignment is empirically observed in the realization of certain utterances, where both manual and non-manual gestures are synchronized with syntactic structures and intonational contours, creating a unified temporal and functional interplay.[4] This is an important empirical argument in favor of our theoretical perspective. To exemplify this idea, we are going to illustrate some examples from our previous work on expressive language, which turns out to be especially interesting to this purpose because, as discussed in Giorgi and Petrocchi (forthcoming), in emotional language we find a higher percentage of co-speech gestures with respect to non-emotional language.

3 Gestures and the sensorimotor component

Research on expressive language has revealed striking cross-linguistic similarities in the non-manual gestures associated with emotions such as surprise and disapproval.[5] Co-speech gestures in surprise – i.e., counter-expectational special questions– and surprise-disapproval questions have been studied in Italian (Giorgi 2016; 2018; Giorgi and Dal Farra 2019, 16 speakers), Spanish (Furlan 2019, 12 speakers), German (Dal Farra, Giorgi and Hinterhölzl, 2018, 8 speakers), Neapolitan (Marchetiello 2022, 16 speakers), Vietnamese (5 speakers), Korean (10 speakers) and Japanese (10 speakers) (Petrocchi 2022). In these works, the same experimental material was used – adapted to the language in question – to make a comparison easier. In all the experiments, a repetition task was proposed. In what follows, we briefly describe the issue.As an exemplification of a counter-expectational questions, consider the following scenario (from Giorgi and Dal Farra, 2019, ex.1):

(4) Scenario: Mary calls me on the phone and tells me that she has a new red dress to wear at tonight's party. When I meet her at the party, I see that she is wearing a blue dress. I'm surprised and say:

(5) Ma non era rosso?
But not was-imp-2s red?
'But wasn't it red?'

Consider now surprise-disapproval questions. A typical scenario is the following (from Giorgi

[4] The idea that speech and gesture are synchronized is not new, as previous research has shown that the stroke of a gesture typically coincides with the main accent of the accompanying utterance. See Kendon (1980), McNeill (1992) Abner et al. (2015) for a review.

[5] Due to space constraints, we are unable to include here a complete discussion the results concerning the similarities across languages. See Giorgi & Petrocchi (2024); in the references we also included the link to the Open Access publication.

and Dal Farra 2019, ex. 2):

(6) Scenario: I see my son with his best trousers kneeling in the dirt in the garden. I think he will ruin his trousers. I am annoyed and utter:

(7) Ma cosa fai?But what do-pres-2s
'But what are you doing?'

These sentences would be infelicitous and even ungrammatical if not accompanied by the correct intonation, which differs from that of canonical questions and crucially contributes to their identification as emotional sentences.[6] Moreover, manual and non-manual co-speech gestures systematically accompany the realization of these sentences across languages. For example, non-manual gestures such as raised – cf. picture 1 – and furrowed eyebrows – cf. picture 2 – are consistently employed in Italian, Spanish, German, Neapolitan, Vietnamese, Korean, Japanese and Italian Sign Language (LIS), see Petrocchi, (2022). These non-manual features frequently align with manual gestures, such as Palm Up Open Hand (PUOH) – cf. picture 3 – in surprise contexts, and iterated artichoke – cf. picture 4 – iterated hands-in-prayer – cf. picture 5 – and iterated PUOH in surprise-disapproval contexts. Note that gestures are not just supplementary in these cases, in that they form an essential part of the grammatical structure. Experimental observations show a systematic alignment between gestures and both prosodic and syntactic components. Specifically, Giorgi & Dal Farra (2019). observed that the stroke phase of the manual gesture consistently aligns with the emphatic pitch in the prosodic contour. This pitch, in turn, corresponds to the nuclear syllable of the verbal form, mainly, or is aligned with negation or the wh-constituent when present. These are the positions that occupy the leftmost syntactic slots available for prosodic prominence within the sentence. Notice that, importantly, the emphatic pitch in emotional utterances is distinct from other types of emphatic pitch, such as those associated with focus.

Giorgi & Dal Farra (2019), Giorgi & Petrocchi (2024) and Petrocchi (2022) investigated the nature of the syntactic trigger for the typical gestures accompanying surprise and surprise disapproval sentences.

[6] For a discussion of their syntactic peculiarities, see Giorgi (2016, 2018). On emotional questions, dubbed *special question*, see among the others Obenauer (2004, 2006), Bayer and Obenauer (2011), Munaro and Obenauer (1999), Munaro and Poletto (2003), Obenauer and Poletto (2000), Hinterhölzl and Munaro (2015) and Vicente (2010).

According to Giorgi (2016, 2018), *ma* (but) in Italian is a *discourse (DIS) head*.[7] This means that it is external to the sentence, signaling adversativity. Again, we cannot reproduce the relevant evidence and discussion here and will just highlight the main points and conclusions. As an exemplification, the representation of a sentence such as *Maria è ricca, ma non è felice* (Maria is rich, but she is not happy), is the following one:

(8) [$_{\text{DIS}}$ [$_{\text{CP}}$ Maria è ricca] [**ma** $_{\text{DIS}}$ [$_{\text{CP}}$ non è felice]]

Maria is rich **but** she is not happy

Moreover, surprise and disapproval sentences are in the scope of an *evaluative (EVAL) head* (see Cinque, 1999), which gives them their peculiar emotional interpretation. The syntactic structure assigned to these sentences in Italian is therefore the following:

(9) [$_{\text{DIS}}$ [CP …] [**ma** $_{\text{DIS}}$ [$_{\text{EVAL}}$ [WH- [ ]]]]]

In emotional adversative sentences, the Specifier position of the Discourse projection is occupied by the *expectations* created in the speaker by the specific contexts, as illustrated in examples (4)-(7) above. Following this analysis, we propose that the emotional interpretation specific to these sentences realized through characteristic prosodic and gestural patterns, arises from a distinctive syntactic representation that remains consistent across languages. Hence, the trigger is provided mainly by the presence of the discourse head, followed by an evaluative head, having a scope on the whole sentence. The presence of these heads in the representation determines the sensorimotor realization, both as prosody and gestures.[8]

Summarizing, from the data currently available, it appears that surprise and surprise-disapproval questions carry somewhat different expressive values, which emerge in a slightly

---

[7] Again, for reasons of space, we cannot provide here an exhaustive analysis of adversative sentences and, in particular, of the comparison between *ma* (but) and *però* (but, however), where *però* can be better characterized as an adverb. There are several syntactic and semantic issues to be considered, and moreover, emotional and non-emotional adversative contexts, as expected, are not realized in exactly the same way with respect to their prosody and gestures. We are addressing further issues concerning adversativity in forthcoming works (Giorgi, forthcoming, and Giorgi & Petrocchi, forthcoming).

[8] Giorgi and Petrocchi (forthcoming) are collecting experimental evidence, by means of a repetition task similar to the one illustrated here, suggesting that adversativity triggers a similar gestural pattern, even in the absence of strong emotional content. However, there are systematic differences from both qualitative and quantitative point of view. Therefore, while the hypothesis proposed in this paper is appropriate, it needs to be better specified to incorporate this new empirical evidence.

different gestural pattern while sharing the same syntactic structure. Specifically, they are introduced by the adversative particle, which exhibits unique properties in these contexts. The same phenomenon is observed in typologically distant languages, such as Romance and Germanic languages, as well as in Japanese, Korean, and Vietnamese, which differ considerably in their prosodic and syntactic systems. This suggests that the discourse head and the evaluative projection may serve across languages as syntactic triggers for the realization of non-manual gestures, such as raised and furrowed eyebrows, and manual ones, such as the PUOH gesture, the artichoke gesture, and the hands-in-prayer gesture, all of which realized with the characteristic iterative movement.

Hence, we can conclude that languages are very similar as far as co-speech gestures are concerned and that, therefore, according to our hypothesis, they share an underlying common abstract representation relevant for the gestural realization.

4 Multimodality in language

Summarizing so far, the alignment of syntax, prosody, and gesture observed cross-linguistically points to the existence of an integrated multimodal framework for language.[9] In outr view, an adequate account of linguistic competence should incorporate the combinatorial structures and sensory integration that characterize a multimodal linguistic expression.

Our experimental investigations revealed that surprise and surprise-disapproval questions in Italian, Spanish, German, Neapolitan, Korean, Vietnamese, and Japanese share interpretive, prosodic, syntactic, and gestural features. The alignment between the gestural, prosodic, and syntactic components indicates that gestures systematically synchronize with syntactic and prosodic triggers, providing compelling evidence for the integration of these modalities.

Given our empirical observations, it is plausible to hypothesize that syntactic structures act as the primary drivers for gestural articulation, setting the stage for prosodic and pragmatic alignment as well. Gestures are not just supplmentary but are intrinsically tied to the syntactic computation, with specific configurations systematically eliciting corresponding manual and non-manual cues. These findings challenge the notion of modality-specific constraints.

In this sense, multimodality in language is understood as the sensory-motor realization of syntactic structures. Specifically, we argue that the trigger for multimodal expression is located within the syntax itself, in particular in dedicated evaluative and adversative heads. Information generated at the core syntactic level is read at the interfaces, where it provides the

[9] For a discussion along slightly different lines, see Cohn and Schilperoord (2024).

necessary commands for both the prosodic and gestural realization of the utterance, as well as for its interpretation. Within this perspective, syntax, prosody, and gesture are not independent modules, but components of an integrated communicative system. Indeed, recent research provides mounting evidence for the prosodic integration of co-speech gestures—e.g. beat gestures—within the speech stream, suggesting a level of coordination that goes beyond mere semantic alignment with individual words. For example, Türk and Calhoun (2023) show that gesture phrases synchronize with prosodic phrases, and that this synchronization persists across units at different levels of their respective hierarchical structures. This suggests that gesture and prosody are more closely coordinated than previously assumed. Leonard and Cummins (2011) further demonstrate that the point of maximum extension in beat gestures aligns most consistently with pitch accents, reinforcing the view that gestures and prosody function as a unified system. Similarly, Ferré and Mettouchi (2020) studied the relation between beat and pragmatic gestures and prosodic units noting that the gestures that can reflect speaker stance align with discourse-level structures; for instance, when a hand flip signals or reinforces grammatical modality, the gesture may be maintained throughout the production of the entire utterance. Finally, Drijvers and Özyürek (2018) show that native listeners benefit more from gestures under degraded auditory conditions, whereas non-native listeners struggle to integrate gesture and speech as efficiently in similar contexts. Together, these findings call for theoretical frameworks that treat gesture, prosody and syntax as jointly organized.

Interestingly, it can be observed that speakers gesture even when there is no visible interlocutor or when manual movements are restricted, compensating by intensifying non-manual cues, such as raised shoulders (Giorgi & Dal Farra, 2019) or an increase in eyebrow gestures (Petrocchi 2022; Giorgi & Petrocchi 2024).[10] This phenomenon indicates that gestures are not merely clarificatory tools for oral communication but are integral to the production process itself. Such findings align with studies showing that gestural behavior persists regardless of visibility, reflecting deep-rooted connections between speech and gestures (McNeill 1992; 2005; Kendon 2004) that are grammatical rather than communicatively driven. These results also agree with the psychological literature suggesting that emotions such as surprise and disapproval do not serve a purely communicative function (Ekman & Oster, 1979; Ekman, 1992; Ekman & Friesen, 1971). Rather than being produced to convey information to others, emotions are expressions of the speaker's internal state (Elliott & Jacobs, 2013), observable

[10] See also the work of Bavelas et al. (2008), which explores the independent roles of dialogic involvement and visibility in gesture production. However, the gestures analyzed in their study are mainly representational or interactional, differing from the gestures examined in our research.

even when the individual is alone. Specifically, Ekman & Oster (1979) and Ekman (1992) proposed that a set of facial expressions is innate and directly tied to the experience of specific emotions; for example, raised eyebrows signify *I feel surprised*. This hypothesis aligns with the broader universality hypothesis in psychological literature, which posits that a basic subset of emotions exists across all cultures (Frijda, 1986).

Furthermore, these so-called basic emotions are expressed in universally similar ways, making them cross-culturally identifiable (Ekman and Oster, 1979; Ekman, 1993; Izard, 1994). For instance, surprise is universally associated with raised eyebrows, while anger is marked by furrowed eyebrows. Our findings support this hypothesis, as these facial expressions have been consistently observed in surprise and surprise-disapproval questions across all the languages we have investigated. Thus, in expressing her emotions of surprise or disapproval/anger, the speaker does not aim to communicate her feelings but rather to experience or *embody* them.

As pointed out above, the manual gestures discussed in this paper fall into the category of co-speech gestures. We analyze co-speech gestures within a framework grounded in Kendon's (1995, 2004) classification of gesture functions. The gestures under investigation fulfill a modal function, as they enable speakers to express a stance—such as certainty, uncertainty, or evaluation—towards their own utterances. This aligns with Müller's (2004) interpretation of such gestures as metaphorical in form and pragmatic in function: they do not depict semantic content directly but instead operate at the level of discourse management or speech act specification. Following Kendon's typology, these gestures belong to the broader class of pragmatic gestures, which are not representational but serve to indicate discourse-related meanings such as speaker attitude or communicative intent. Importantly, we adopt Kendon's inclusive view of co-speech gesture, which encompasses both manual and non-manual forms (cf. Kendon 2004; see also Bavelas et al. 2022). The spontanepous gestures we observed do not replace speech but instead synchronize with it. This indicates that both modalities, i.e., speech and gesture, convey a single, unified meaning co-expressively: speech and co-speech gestures jointly express the associated meaning, achieving a multilinear integration that is realized synchronically. They align with the leftmost syntactic slot available when the discourse head DIS° is omitted. Adversativity does not necessarily coexist with emotional language. Adversative sentences can, but must not, be enriched with emotional nuances such as surprise or surprise- disapproval. When emotional components are present, however, gestures are produced with greater amplitude, faster execution, and an increased number of iterations – in the case of surprise-disapproval – whereas gestures associated with non-disapproval surprise exhibit no iteration. Building on the traditional linguistic hypothesis that posits a relationship between language and its interfaces, such as auditory and motor-articulatory systems, at the periphery of language (Chomsky, 2000; Hauser et al., 2002), our findings lead us to expand our framework by incorporating the most recent evidence on the importance of visual, auditory, and gestural cues in the construction of utterances (Cohn & Foulsham, 2022; Hagoort & Özyürek, 2024; Holler & Levinson, 2019). Notably, recent studies have examined how multimodal cues contribute to inferring intended meanings (Bettelli & Panzeri, 2023; Canal et al., 2022; Domaneschi et al., 2017) and have demonstrated the existence of multiple pathways – ranging from purely linguistic to sensory, experiential, and social – for representing the meanings of lexical items (Frau et al., 2024; Pexman, 2020) and entire utterances, including abstract and figurative expressions (e.g., Carston, 2010; Cuccio, 2022; Tettamanti et al., 2005). Indeed, the idea that the faculty of language and its performance mechanisms must be precisely located where they have been biologically assigned – namely, in the human body (the brain) and at the interface with the pragmatic environmental context – aligns with the minimalist approach.

Thus, moving beyond the classic model of language, which treats language as the amodal translation of sensorimotor experience into a set of representations and computations independently processed (Fodor, 1987; 1983; Pylyshyn, 1984), as well as the recent modal-specific theories of multimodal language within the embodied cognition framework (Borghi and Cimatti, 2010), we align with a more efficient perspective. This perspective suggests that linguistic representations are organized across multiple dimensions, ranging from amodal linguistic information – such as abstract semantic feature lists and semantic networks – to experience-based dimensions – including sensory-motor, emotional, and social experiences. As far as we can observe, much of what pertains to grammar and syntax in the strict sense remains detached from purely *embodied* explanations (Frau et al., 2024), as it continues to reside within the domain of linguistic arbitrariness (Monaghan et al., 2014). However, linguistic computations are not merely cognitive processes, in that they are deeply integrated with embodied experiences and social interactions.

5 Conclusions

As exemplified in the previous sections, the presence of the adversative discourse head is critical, as evidenced by its consistent identification across all the languages studied. Consequently, our hypothesis is that DIS triggers the *type* of gesture involved across the studied languages. This trigger is manifested in the manual configuration, which serves as a *radical element.* Furthermore, the effect of the EVAL head is evident in the *movement morpheme*, characterized by increased speed in the iterative motion.

The concepts of *gesture-radicals* and *movement morphemes* were first proposed by Isabella Poggi in her studies on quotable gestures in Italian, published in the 1980s (Poggi 1983). She categorized the gesture into these components, drawing on linguistic frameworks inspired by sign language research. Poggi systematically analyzed these gestures from semantic and grammatical perspectives, explicitly comparing them with lexical units in spoken Italian. Consider, for instance, the artichoke gesture, also referred to as *Hand Pursue – Mano a Borsa*, or *Mano a Tulipano*. This gesture was initially described by Poggi (1983) as consisting of the gestural radical, i.e. the *purse hand* (a pouch-like handshape) extended forward at the forearm, combined with the slow iterative up-and-down motion. Poggi also describes this gesture as being usually associated with pseudo-questions, particularly rhetorical questions, expressing criticism toward the interlocutor. As Kendon (1992) later highlighted, Poggi's examples of pseudo-questions included structures such as *What are you doing?* which intriguingly aligns with some of the constructions analyzed in our study. According to Poggi, this gesture is often accompanied by *knitted eyebrows*.

Based on our observations, when the emotional content – disapproval - is activated, this gesture undergoes a transformation characterized by faster up-and-down movements and a higher frequency of iterations. Adversativity thus functions as a syntactic trigger for gestures such as PUOH, the artichoke gesture, and the hands-in-prayer gesture, all of which share the iterative, unmarked slow movement pattern. On the other hand, emotional components add nuances to these gestures, appropriate for the emotional contexts, influencing the movement morphemes. Indeed, in the case of the adversative surprise counter-expectational questions, the radical gesture can remain the same as in the surprise-disapproval contexts (e.g., the PUOH gesture), but the iterative movement is abruptly halted. These enhancements underscore the interplay between adversative and evaluative interpretive layers, reinforcing the hypothesis of a grammatical trigger for gestural realizations across the studied languages.

Finally, our findings extend Poggi's early observations by uncovering further variations in amplitude and frequency in gestures, which appear to correlate directly with the presence of the head EVAL.

In future research, we will further develop this idea by experimentally investigating how emotional contexts differ from non-emotional ones, while comparing various syntactic structures and checking the syntax-prosody-gesture alignment.

References

Abner, N., Cooperrider, K., & Goldin-Meadow, S. (2015). Gesture for linguists: A handy primer. *Language and Linguistics Compass*, 9 (11), 437-449.

Avesani, C. (1995). ToBIt: Un sistema di trascrizione per l'intonazione italiana. In G. Lazzari (Ed.), *Atti delle V Giornate di Studio del Gruppo di Fonetica Sperimentale dell'Associazione Italiana di Acustica (A.I.A)* (pp. 85-98). Trento: Servizio Editoria ITC. Bayer, J., & Obenauer, H.-G. (2011). Discourse particles, clause structure, and question types. *The Linguistic Review*, 28(4), 449-491.

Bavelas, J. (2022) *Face-to-face dialogue: theory, research, and applications*. Oxford, UK: Oxford University Press.

Bavelas, J. B., Gerwing, J., Sutton, C., & Prevost, D. (2008). Gesturing on the telephone: Independent effects of dialogue and visibility. *Journal of Memory and Language, 58*(2), 495–520. https://doi.org/10.1016/j.jml.2007.02.004.

Bettelli, G. & Panzeri, F. (2023). Irony is easy to understand: The role of emoji in irony detection. *Intercultural Pragmatics*, 20(5), 467-493. https://doi.org/10.1515/ip-2023-5001.

Borghi, A. M. & Cimatti, F. (2010). Embodied cognition and beyond: Acting and sensing the body. *Neuropsychologia*, 48, 763-773.

Canal, P., Bischetti, L., Bertini, C., Ricci, I., Lecce, S., Bambini, V. (2022). N400 differences between physical and mental metaphors: The role of Theories of Mind. *Brain and Cognition*, 161, 105879. https://doi.org/10.1016/j.bandc.2022.105879.

Carston, R. (2010). Metaphor: Ad hoc concepts, literal meaning and mental images. *Proceedings of the Aristotelian Society*, 110(3), 295-321. https://doi.org/10.1111/j.1467-9264.2010.00288.x.

Chomsky, N., Gallego A., Ott D. (2019) Generative grammar and the faculty of language: Insights, questions, and challenges. Catalan Journal of Linguistics 2019(Special Issue): 229–261. https://doi.org/10.5565/rev/catjl.288.

Chomsky, N. (1957). *Syntactic Structures*. The Hague: Mouton.

Chomsky, N. (1975). *Reflections on Language*. New York: Pantheon Books.

Chomsky, N. (1995). *The Minimalist Program*. Cambridge, MA: MIT Press.

Chomsky, N. (2000). Minimalist inquiries: The framework. In R. Martin, D. Michaels, & J. Uriagereka (Eds.), *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik* (pp. 89-155). Cambridge, MA: MIT Press.

Chomsky, N. (2001). Derivation by phase. In M. Kenstowicz (Ed.), *Ken Hale: A Life in Language* (pp. 1-52). Cambridge, MA: MIT Press.

Chomsky, N. (2008). On phases. In R. Freidin, C. P. Otero, & M. L. Zubizarreta (Eds.), *Foundational Issues in Linguistic Theory: Essays in Honor of Jean-Roger Vergnaud* (pp. 133-166). Cambridge, MA: MIT Press.

Cinque, G. (1999). *Adverbs and Functional Heads*. Oxford: Oxford University Press.

Cohn, N. & Schilperoord, J. (2024). *A multimodal Language Faculty: A cognitive framework for communication*, London: Bloomsbury.

Cohn, N. & Foulsham, T. (2022). Meaning above (and in) the head: Combinatorial visual morphology from comics and emoji. *Memory & Cognition*, 50(7), 1381-1398. https://doi.org/10.3758/s13421-022-01294-2.

Colasanti, V. (2021a). Linguistically-integrated gestures: A case study from the languages of Southern Italy. Paper presented at the *Annual Meeting of the LAGB*, 6-9 September 2021.

Colasanti, V. (2021b). Syntactically-integrated gestures in Southern Italy. Paper presented at *Going Romance 35*, University of Amsterdam, 1-3 December 2021.

Colasanti, V. (2023). Functional gestures as morphemes: Some evidence from the languages of Southern Italy. *Glossa: A Journal of General Linguistics*, 8(1).

Cuccio, V. (2022). Pragmatica ed *embodiment*. In F. Domaneschi & V. Bambini (Eds.), *Pragmatica sperimentale* (pp. 237-246). Bologna: Il Mulino.

D'Anselmo, A., Prete G., Zdybek P.,Tommasi L. and Brancucci A. (2019), Guessing Meaning From WorSounds of Unfamiliar Languages: A Cross-Cultural Sound Symbolism Study. *Frontiers in Psychology* 10:593.doi: 10.3389/fpsyg.2019.00593.

Dal Farra, C., Giorgi, A., & Hinterhölzl, R. (2018). Syntax, prosody and gesture in expressing surprise. Paper presented at the 40th Annual Conference of the German Society of Linguistics, *Workshop on Evaluative Meanings: Theoretical and Computational Perspectives*, 7 March 2018, Stuttgart.

De Saussure, F. (1916). *Course in General Linguistics*. New York: McGraw-Hill.

Domaneschi, F., Passarelli, M., & Chiorri, C. (2017). Facial expressions and speech acts: Experimental evidences on the role of the upper face as an illocutionary force indicating device in language comprehension. *Cognitive Processing*, 18(3), 285-306. https://doi.org/10.1007/s10339-017-0809-6.

Drijvers, L., & Özyürek, A. (2018). Native language status of the listener modulates integration of speech and iconic gestures in clear and degraded speech. *Journal of Memory and Language, 177-178*, 7-17. https://doi.org/10.1016/j.bandl.2018.01.003.

Ebert, C., & Ebert, C. (2014). Gestures, demonstratives, and the attributive/referential distinction. Paper presented at *Semantics and Philosophy in Europe (SPE 7)*, Humboldt-Universität zu Berlin.

Ekman, P. (1973). Universal facial expressions in emotion. *Studia Psychologica*, 15(2), 140- 147.

Ekman, P., & Oster, H. (1979). Facial expressions of emotion. *Annual Review of Psychology*, 30, 527-554.

Ekman, P. (1992). An argument for basic emotions. *Cognition & Emotion*, 6(3-4), 169-200.

Ekman, P. (1993). Facial expression and emotion. *American Psychologist*, 48(4), 384–392. https://doi.org/10.1037/0003-066X.48.4.384.

Ekman, P., & Friesen, W. V. (1978). *Facial Action Coding System: A Technique for the Measurement of Facial Movement*. Palo Alto: Consulting Psychologists Press.

Esipova, M. (2019). Acceptability of at-issue co-speech gestures under contrastive focus. *Glossa: A Journal of General Linguistics*, 4(1), 11. https://doi.org/10.5334/gjgl.635.

Ferré, G. (2012). Functions of three open-palm hand gestures. *Multimodal Communication* 1(1), 5-20. 10.1515/mc-2012-0002.

Ferré, G., & Mettouchi, A. (2020). A cultural study of open-palm hand gestures and their prosodic correlates. *Proceedings of the 10th International Conference on Speech Prosody 2020* (pp. 285–289). Tokyo, Japan. https://doi.org/10.21437/speechprosody.2020-58.

Fodor, J. A. (1983). *The Modularity of Mind: An Essay on Faculty Psychology*. Cambridge: MIT Press.

Fodor, J. A. (1987). *Psychosemantics: The Problem of Meaning in the Philosophy of Mind*. Cambridge: MIT Press.

Frijda, N. H. (1986). *The Emotions*. Cambridge: Cambridge University Press.

Furlan, E. (2019). *Preguntas especiales en contextos de sorpresa: Entonación y gestualidad en castellano. Análisis de datos experimentales procedentes de entrevistas a nativos*. MA Thesis. (Supervisor A.Giorgi), Ca’ Foscari University.

Giorgi, A. (2014). Prosodic Signals as Syntactic Formatives in the Left Periphery, in A. Cardinaletti et al. (eds.) *On Peripheries: Exploring Clause Initial and Clause Final Positions*, Tokyo, Hituzi Syobo Press, 161-188.

Giorgi, A. (2015). Discourse and the syntax of the left periphery. In J. Bayer, R. Hinterhölzl, & A. Trotzke (Eds.), *Discourse-oriented Syntax* (pp. 229-250). New York: John Benjamins.

Giorgi A. (2016), On the temporal interpretation of certain surprise questions. *SpringerPlus*. 5(1390). 1–10.

Giorgi A. (2018), Ma non era rosso? (But wasn't it red?): On counter-expectational questions in Italian, L. Repetti, F. Ordóñez (eds.). *Selected papers from the 46th Linguistic Symposium on Romance Languages (LSRL),* New York, John Benjamins, 69-84.

Giorgi, A. (2023b). Pragmatics in the Minimalist framework: Evidence from the study of emotional language. *Evolutionary Linguistic Theory*, 5(2), 103–127.

Giorgi, A. (forthcoming), On the syntax and pragmatics of adversativity in Italian, Ca' Foscari University of Venice.

Giorgi, A. & C. Dal Farra (2019). On the Syntax/Pragmatics Interface: Expressing Surprise and Disapproval. *Intercultural Pragmatics*, 16(3), 335-361. https://doi.org/10.1515/ip-2019-0017.

Giorgi, A. & E. Petrocchi (2024). A cross-cultural analysis of the gestural pattern of surprise and surprise-disapproval questions. *Intercultural Pragmatics*, 21(3), 307–347. https://doi.org/10.1515/ip-2024-3002.

Giorgi, A. & E. Petrocchi (forthcoming). The gestures of adversativity, Ca' Foscari University of Venice.

Grice, M., M. D'Imperio, M. Savino & C. Avesani (2005). Strategies for Intonation Labelling across Varieties of Italian. In S.-A. Jun (ed.), *Prosodic Typology*, 362-389. Oxford: Oxford University Press.

Gupta, S., E. Pequay, C. François & I. Dautriche (2024). Forms and Functions of Gestures in Preverbal 12- to 15 Months Old Infants. *Infancy*, 30, e12645. https://doi.org/10.1111/infa.12645.

Hauser, M. D., N. Chomsky & T. W. Fitch (2002). The Faculty of Language: What Is It, Who Has It, and How Did It Evolve? *Science*, 298(5598), 1569–1579. https://doi.org/10.1126/science.298.5598.1569.

Hagoort, P. & A. Özyürek (2024). Extending the Architecture of Language From a Multimodal Perspective. *Topics in Cognitive Science*. https://doi.org/10.1111/tops.12728.

Holler, J. & S. C. Levinson (2019). Multimodal Language Processing in Human Communication. *Trends in Cognitive Sciences*, 23(8), 639–652. https://doi.org/10.1016/j.tics.2019.05.006.

Imai, M., Kita, S., Nagumo, M., and Okada, H. (2008). Sound symbolism facilitates early verb learning. *Cognition*, 109, 54–65. doi: 10.1016/j.cognition.2008.07.015.

Holle, H., Obermeier, C., Schmidt-Kassow, M., Friederici, A. D., Ward, J., & Gunter, T. C. (2012). Gesture facilitates the syntactic analysis of speech. *Frontiers in Psychology*, 3, 74.

https://doi.org/10.3389/fpsyg.2012.00074
Ippolito, M. (2019). Gestures as markers on non-canonical questions. In *Proceedings of the 22nd Amsterdam Colloquium*, 574-583.
Ippolito, M. (2021). The Contribution of Gestures to the Semantics of Non-Canonical Questions. *Journal of Semantics*, 38(3), 363–392. https://doi.org/10.1093/jos/ffab007.
Ippolito, M., F. Foppolo & F. Panzeri (2022). Is the Mano a Tulipano gesture compatible with canonical questions? An empirical study of a speech act marking gesture. In *Proceedings of Sinn und Bedeutung*, 451–464.
Izard, C. E. (1994). Innate and universal facial expressions: Evidence from developmental and cross-cultural research. *Psychological Bulletin*, 115(2), 288–299. https://doi.org/10.1037/0033-2909.115.2.288.
Jouitteau, M. (2004). Gestures as expletives: multichannel syntax. In G. Garding & M. Trujimura (eds.), *WCCFL 23 Proceedings*, 101–114. Somerville, MA: Cascadilla Press.
Jouitteau, M. (2007). Listen to the sound of salience: Multichannel syntax of Q particles. In Sergio Baauw & Frank Drijkoningen (eds.), *Romance Languages and Linguistic Theory 2007*. Amsterdam: John Benjamins.
Kantartzis, K., Imai, M., and Kita, S. (2011). Japanese sound-symbolism facilitates word learning in English-speaking children. *Cognitive Science*, 35, 575–586. doi: 10.1111/j.1551-6709.2010.01169.x.
Kendon, A. (1980). Gesticulation and speech: Two aspects of the process of utterance. In Mary R. Key (ed.), *The relationship of verbal and nonverbal communication*, 207-227. The Hague: Mouton.
Kendon, A. (1992). Some Recent Work from Italy on Quotable Gestures (Emblems). *Journal of Linguistic Anthropology* 2(1), 92-108.
Kendon, A. (2004). *Visible Action as Utterance*. Cambridge: Cambridge University Press.
Monaghan, P., Richard S., Morten H. C. & Simon K. (2014). How arbitrary is language? *Philosophical Transactions of the Royal Society B: Biological Sciences*, 369, 20130299. https://doi.org/10.1098/rstb.2013.0299.
Leonard, T., & Cummins, F. (2011). The temporal relation between beat gestures and speech. *Language and Cognitive Processes, 26*(10), 1457–1471. https://doi.org/10.1080/01690965.2010.500218
Marchetiello, C. (2022). *Uè, ma che staje facenno? (Hey, but what are you doing?): Expressing Expectation and Surprise/Disapproval in Neapolitan Dialect*. MA Thesis. Venice: Ca' Foscari University.

McNeill, D. (1992). *Hand and Mind: What Gestures Reveal about Thought*. Chicago: University of Chicago Press.

McNeill, D. (2005). *Gesture and Thought*. Chicago: University of Chicago Press. https://doi.org/10.7208/chicago/9780226514642.001.0001.

McNeill, D. (2014). *Why We Gesture: The Surprising Role of Hand Movements in Communication*. Cambridge: Cambridge University Press.

Müller, C. (2004). Forms and uses of the Palm Up Open Hand: A case of a gesture family? In C. Müller & R. Posner (Eds.), *The semantics and pragmatics of everyday gestures* (pp. 233–256). Weidler.

Munaro, N. & Obenauer H.G. (1999). On underspecified wh-elements in pseudo-interrogatives. *University of Venice Working Papers in Linguistics*, 9(1-2), 181-253.

Munaro, N. & Poletto C. (2003). Sentential particles and clausal typing in the Veneto dialects. *University of Venice Working Papers in Linguistics*, 13, 127-154.

Obenauer, H.G. (2004). Nonstandard wh-questions and alternative checkers. In L. Lohnstein & S. Trissler (eds.), *Syntax and Semantics of the Left Periphery*, 343-384. Berlin: Mouton de Gruyter.

Obenauer, H.G. (2006). Special interrogatives, left-periphery, wh-doubling and (apparently) optional elements. In Doetjes & González (eds.), *Selected Papers from Going Romance 2004*, 247-273. Amsterdam: John Benjamins.

Obenauer, H.G. & Poletto C. (2000). Rhetorical wh-phrases in the left periphery of the sentence. *Working Papers in Linguistics, University of Venice*, 10, 121-151.

Petrocchi, E. (2022). *Multimodality in Grammar: The Case of Special Questions Expressing Surprise in Oral and Sign Language*. PhD Thesis. Venice: Ca' Foscari University.

Pexman, P. M. (2020). How does meaning come to mind? Four broad principles of semantic processing. *Canadian Journal of Experimental Psychology*, 74(4), 275-283. https://doi.org/10.1037/cep0000235.

Poggi, I. (1983). La mano a borsa: Analisi semantica di un gesto emblematico olofrastico. In G. Attili & P. E. Ricci Bitti (eds.), *Comunicare senza parole*, 219-238. Rome: Bulzoni.

Pylyshyn, Z. (1984). *Computation and Cognition: Toward a Foundation for Cognitive Science*. Cambridge: MIT Press.

Sailor, C. & Colasanti C. (2020). Co-speech gestures under ellipsis: A first look. Paper presented at the *2020 LSA Annual Meeting*.

Schlenker, P. (2018). Gesture projection and cosuppositions. *Linguistics and Philosophy*, 41, 295–365. https://doi.org/10.1007/s10988-017-9225-8.

Tettamanti, M., Buccino G., Saccuman M.C., Gallese V., Danna M., Scifo P., Fazio F. Rizzolatti G., Cappa S. F. & Perani D. (2005). Listening to action-related sentences activates fronto-parietal motor circuits. *Journal of Cognitive Neuroscience*, 17(2), 273–281. https://doi.org/10.1162/0898929053124965.

Türk, O., & Calhoun, S. (2023). Phrasal Synchronization of Gesture with Prosody and Information Structure. *Language and Speech*, 67(3), 702-743. https://doi.org/10.1177/00238309231185308 .

Vicente, L. (2010). On the Syntax of Adversative Coordination. *Natural Language and Linguistic Theory*, 28(2), 381-415.

Zhang, Y., Frassinelli, D., Tuomainen, J., Skipper, J. I., & Vigliocco, G. (2021). More than words: word predictability, prosody, gesture and mouth movements in natural language comprehension. Proceedings. *Biological sciences*, 288(1955), 20210500. https://doi.org/10.1098/rspb.2021.0500 .

Pictures:

Picture 1: Raised eyebrows

Picture 2: Furrowed eyebrows

Picture 3: PUOH

Picture 4: Hands in prayer

Picture 4: Artichoke